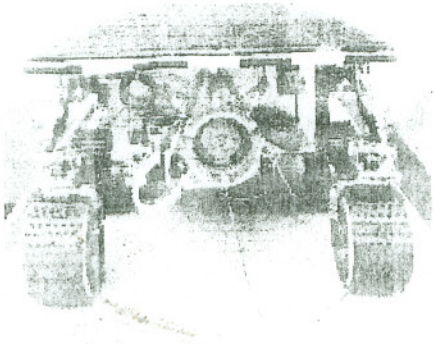
ANALYTICAL CUMULATIVE EXAMINATION
15th November 1997

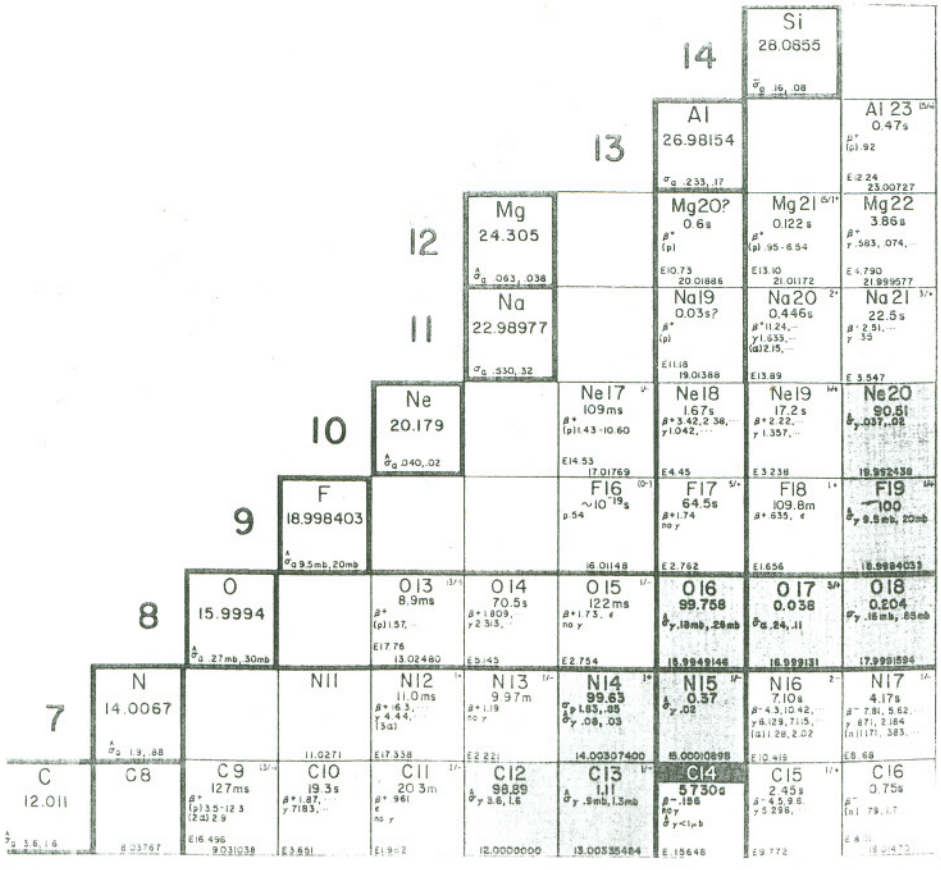
The Mars Pathfinder Alpha Proton X-Ray Spectrometer



This instrument is a foreign-provided derivative of instruments flown the Russian Vega and Phobos missions and identical to the unit that was planned for flight on the Russian Mars '96 missions. Accordingly, the instrument has extensive, applicable flight heritage. With the mobility provided by the microrover and a deployment mechanism, the APX Spectrometer not only acquires spectra from the ubiquitous martian dust, but more importantly, is deployed to distinct rock outcropping, thereby analyzing the native rock composition for the first time. The alpha and proton spectrometer portions are provided by the Max Planck Institute, Department of Chemistry, Mainz Germany. The x-ray spectrometer portion is provided by the University Of Chicago.

This elemental composition instrument consists of alpha particle source and detectors for back-scattered alpha particles, protons and X-Rays. The APX Spectrometer will determine elemental chemistry of surface materials for most major elements except hydrogen. The analytical process is based on three interactions of alpha particles with matter: elastic scattering of alpha particles by nuclei, alpha-proton nuclear reactions with certain light elements, and excitation of the atomic structure of atoms by alpha particles, leading to the emission of characteristic X-rays. The approach used is to expose material to a radioactive source that produces alpha particles with a known energy, and to acquire energy spectra of the alpha particles, protons and X-Rays returned from the sample. Such an instrument can identify and determine the amounts of most chemical elements. The basis of the alpha mode of the instrument is the dependence of the energy spectrum of alpha particles scattered from a surface on the composition of the surface material. The method has the best resolving power for the lighter elements.

The proton spectra for alpha particles interacting with elements with atomic numbers from 9 to 14 are very characteristic of the individual elements, reflecting the resonance nature of the nuclear interactions involved. The proton mode allows their detection and measurement. The addition of a third detector for X-Rays Results in a significant extension of the accuracy and sensitivity of the instrument, particularly for the heavier, less abundant elements. Alpha sources produce characteristic X-rays for a range of elements, giving an instrument sensitivity that can approach the ppm level. The APXS sensor head is mounted external to the Rover chassis on a deployment mechanism (described below). This mechanism places the APXS in contact with rock and soil surfaces. The APXS electronics are mounted within the rover, in a temperature-controlled environment.

The above is an excert from a document describing the science instruments on the Mars Pathfinder spacecraft that landed on Mars on July 4th 1997. Write an essay discussing the following:

1. (10%) The nature of the α source likely to have been used in the APX.
2. (40%) The nature of the (α,p) reactions likely to have been used to analyze for light elements. (An excerpt from the Chart of the Nuclides is attached that you might useful).
3. (10%) The processes giving rise to the X-rays.
4. (40%) The detectors used for the scattered α particles, the protons and the X-rays.